

CHRISTMAS ISLAND, Christmas IslandWMO: **969950**Lat: **10.4516S**Lon: **105.6890E**Elev: **860**StdP: **14.25**Time Zone: **7.00 (E07)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	69.5	70.5	61.4	83.9	74.0	63.0	88.9	74.0	20.4	75.2	19.1	75.5	5.8	130	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	6.6	84.0	77.0	82.9	76.3	82.5	76.1	79.1	81.2	78.2	80.7	77.7	80.3	9.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	154.2	80.0	77.4	147.0	79.6	77.0	145.0	79.4	43.4	80.6	42.5	81.6	41.8	80.4	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	17.4	16.4	DB	67.1	85.8	1.4	2.0	66.1	87.3	65.2	88.5	64.4	89.6	63.4	91.1	(4)
(5)			WB	63.6	80.7	2.0	2.4	62.1	82.4	61.0	83.8	59.8	85.1	58.4	86.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	77.4	77.5	78.0	78.6	78.3	77.3	75.6	75.0	75.3	76.3	76.9	77.1
	DBStd	1.84	1.55	1.66	1.61	1.54	1.51	1.36	1.42	1.43	1.12	1.13	1.46	1.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9832	849	771	869	857	878	818	794	776	759	814	807	840
	CDD65	4357	384	351	404	407	413	368	329	311	309	349	357	375
	CDH74	24114	2415	2252	2797	3111	3052	2117	1163	858	863	1398	1872	2216
	CDH80	1488	227	220	309	270	173	40	7	5	4	18	66	149
Wind	WSAvg	9.9	6.6	6.6	6.7	8.8	10.4	11.4	12.6	12.9	13.3	12.1	10.1	7.1
Precipitation	PrecAvg	72.8	12.0	13.4	11.7	8.3	6.7	6.9	3.0	1.8	1.1	1.4	6.1	8.1
	PrecMax	116.3	25.9	31.9	27.4	22.4	20.8	27.9	14.0	6.5	4.2	4.6	20.7	29.9
	PrecMin	42.0	1.8	2.4	1.4	0.7	0.6	0.4	0.2	0.1	0.0	0.0	0.2	0.5
	PrecStd	22.6	5.9	7.6	6.2	5.7	4.8	7.5	3.1	1.8	0.9	1.5	5.8	7.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.4	84.4	84.6	84.4	83.7	82.3	80.9	80.7	80.6	81.4	82.7	84.1
		MCWB	76.5	76.8	78.1	77.5	77.7	76.1	75.3	74.9	73.8	74.5	74.7	75.7
	2%	DB	82.8	83.2	83.7	83.0	82.5	80.9	79.4	79.1	79.1	80.5	81.6	82.6
		MCWB	75.5	76.2	77.5	76.8	76.9	75.1	73.8	72.7	72.6	73.7	74.4	75.3
	5%	DB	82.3	82.4	82.7	82.4	81.7	80.4	78.8	78.5	78.5	79.4	80.7	81.6
		MCWB	75.2	75.8	76.7	76.5	76.3	74.8	73.3	72.2	72.3	73.2	74.1	74.9
	10%	DB	81.2	81.4	82.0	81.6	80.9	79.4	78.1	77.4	77.4	78.8	79.9	80.7
		MCWB	74.9	75.5	76.4	76.3	75.7	74.4	73.0	71.8	72.1	73.0	73.8	74.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	78.3	82.3	79.5	80.7	78.6	78.4	78.5	77.4	77.2	78.1	78.2
		MCDB	81.8	81.4	83.0	81.9	81.8	79.6	79.0	78.8	77.9	78.8	79.9	80.6
	2%	WB	77.6	77.6	78.5	78.3	78.6	77.2	76.6	76.1	75.6	76.1	76.7	77.3
		MCDB	80.7	80.8	81.4	81.1	80.7	79.1	78.1	77.4	77.2	78.1	78.9	80.0
	5%	WB	76.8	77.1	77.8	77.9	77.7	76.4	75.5	74.3	74.4	75.3	76.1	76.5
		MCDB	80.0	80.3	80.8	80.7	80.1	78.6	77.2	76.4	76.6	77.5	78.4	79.4
	10%	WB	76.3	76.5	77.1	77.3	77.0	75.8	74.4	73.0	73.4	74.6	75.3	76.0
		MCDB	79.4	79.7	80.3	80.1	79.5	78.2	76.6	75.8	75.9	77.1	77.9	78.9
Mean Daily Temperature Range	5% DB	MDBR	8.2	7.8	7.8	6.6	5.7	5.0	5.1	5.6	5.7	6.0	6.5	7.9
		MCDBR	9.0	8.8	8.7	7.0	6.1	5.3	5.5	5.9	6.1	6.5	7.2	8.7
		MCWBR	4.2	4.0	4.7	3.9	3.6	3.3	3.5	3.5	3.4	3.3	3.4	3.9
	5% WB	MCDBR	7.8	7.6	7.8	6.6	5.8	5.1	5.0	5.4	5.5	5.7	6.4	7.6
MCWBR		4.5	4.1	4.9	4.2	4.0	3.5	4.0	4.2	4.3	3.7	3.9	4.1	
Clear-Sky Solar Irradiance	taub	0.390	0.399	0.397	0.404	0.412	0.407	0.397	0.394	0.416	0.449	0.450	0.406	
	taud	2.572	2.556	2.564	2.541	2.504	2.504	2.506	2.490	2.411	2.310	2.340	2.514	
	Ebn at Noon	302	298	293	281	269	266	271	280	283	279	282	296	
	Edh at Noon	34	34	33	33	32	31	32	34	38	43	43	36	
All-Sky Solar Radiation	RadAvg	1980	1957	1859	1667	1593	1483	1593	1814	1974	2081	1999	1990	
	RadStd	209	195	157	148	115	141	111	51	141	175	187	209	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.50	+0.94	N/A	N/A	+0.79	+1.42	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page